CHAPTER 15—MULTIPLE REGRESSION

MULTIPLE CHOICE

1. The mathematical equation relating the expected value of the dependent variable to the value of the independent variables, which has the form of E(y) = $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$ is
   a. a simple linear regression model
   b. a multiple nonlinear regression model
   c. an estimated multiple regression equation
   d. a multiple regression equation

2. The estimate of the multiple regression equation based on the sample data, which has the form of E(y) = $\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_p x_p$ is
   a. a simple linear regression model
   b. a multiple nonlinear regression model
   c. an estimated multiple regression equation
   d. a multiple regression equation

3. The mathematical equation that explains how the dependent variable y is related to several independent variables x_1, x_2, …, x_p and the error term ε is
   a. a simple nonlinear regression model
   b. a multiple regression model
   c. an estimated multiple regression equation
   d. a multiple regression equation

4. A measure of the effect of an unusual x value on the regression results is called
   a. Cook’s D
   b. Leverage
   c. odd ratio
   d. unusual regression

5. A variable that cannot be measured in terms of how much or how many but instead is assigned values to represent categories is called
   a. an interaction
   b. a constant variable
   c. a category variable
   d. a qualitative variable

6. In order to test for the significance of a regression model involving 3 independent variables and 47 observations, the numerator and denominator degrees of freedom (respectively) for the critical value of F are
   a. 47 and 3
   b. 3 and 47
   c. 2 and 43
   d. 3 and 43

7. In regression analysis, an outlier is an observation whose
   a. mean is larger than the standard deviation
   b. residual is zero
   c. mean is zero
   d. residual is much larger than the rest of the residual values

8. A variable that takes on the values of 0 or 1 and is used to incorporate the effect of qualitative variables in a regression model is called
   a. an interaction
   b. a constant variable
   c. a dummy variable
   d. None of these alternatives is correct.

9. In a multiple regression model, the error term ε is assumed to be a random variable with a mean of
   a. zero
   b. -1
   c. 1
   d. any value

10. In regression analysis, the response variable is the
   a. independent variable
   b. dependent variable
   c. slope of the regression function
   d. intercept

11. A multiple regression model has
   a. only one independent variable
   b. more than one dependent variable
   c. more than one independent variable
   d. at least 2 dependent variables

12. A measure of goodness of fit for the estimated regression equation is the
   a. multiple coefficient of determination
   b. mean square due to error
   c. mean square due to regression
   d. sample size

13. The numerical value of the coefficient of determination
   a. is always larger than the coefficient of correlation
   b. is always smaller than the coefficient of correlation
   c. is negative if the coefficient of determination is negative
   d. can be larger or smaller than the coefficient of correlation

14. The adjusted multiple coefficient of determination is adjusted for
   a. the number of dependent variables
   b. the number of independent variables
   c. the number of equations
   d. detrimental situations

15. In a multiple regression model, the variance of the error term ε is assumed to be
   a. the same for all values of the dependent variable
   b. zero
   c. the same for all values of the independent variable
   d. -1

16. In multiple regression analysis, the correlation among the independent variables is termed
   a. homoscedasticity
   b. linearity
   c. multicollinearity
   d. adjusted coefficient of determination

17. In a multiple regression model, the values of the error term ,ε, are assumed to be
    a. zero
    b. dependent on each other
    c. independent of each other
    d. always negative

18. In a multiple regression model, the error term ε is assumed to
    a. have a mean of 1
    b. have a variance of zero
    c. have a standard deviation of 1
    d. be normally distributed

19. In multiple regression analysis,
    a. there can be any number of dependent variables but only one independent variable
    b. there must be only one independent variable
    c. the coefficient of determination must be larger than 1
    d. there can be several independent variables, but only one dependent variable

20. A regression model in which more than one independent variable is used to predict the dependent variable is called
    a. a simple linear regression model
    b. a multiple regression model
    c. an independent model
    d. None of these alternatives is correct.

21. A term used to describe the case when the independent variables in a multiple regression model are correlated is
    a. regression
    b. correlation
    c. multicollinearity
    d. None of the above answers is correct.

22. A variable that cannot be measured in numerical terms is called
    a. a nonmeasurable random variable
    b. a constant variable
    c. a dependent variable
    d. a qualitative variable

23. In logistic regression,
    a. there can only be two independent variables
    b. there are two dependent variables
    c. the dependent variable only assumes two discrete values
    d. the dependent variable only assumes two continuous values

24. In a situation where the dependent variable can assume only one of the two possible discrete values,
    a. we must use multiple regression
    b. there can only be two independent variables
    c. logistic regression should be applied
    d. all the independent variables must have values of either zero or one

25. In a regression model involving more than one independent variable, which of the following tests must be used in order to determine if the relationship between the dependent variable and the set of independent variables is significant?
    a. t test
    b. F test

c. Either a t test or a chi-square test can be used.
d. chi-square test

26. For a multiple regression model, SSR = 600 and SSE = 200. The multiple coefficient of determination is
a. 0.333
b. 0.275
c. 0.300
d. 0.75

27. In a multiple regression analysis involving 15 independent variables and 200 observations, SST = 800 and SSE = 240. The coefficient of determination is
a. 0.300
b. 0.192
c. 0.500
d. 0.700

28. A regression model involved 5 independent variables and 136 observations. The critical value of t for testing the significance of each of the independent variable's coefficients will have
a. 121 degrees of freedom
b. 135 degrees of freedom
c. 130 degrees of freedom
d. 4 degrees of freedom

29. The multiple coefficient of determination is
a. MSR/MST
b. MSR/MSE
c. SSR/SST
d. SSE/SSR

30. A multiple regression model has the form
$\hat{y}=7+2x_1+9x_2$
As x_1 increases by 1 unit (holding x_2 constant), y is expected to
a. increase by 9 units
b. decrease by 9 units
c. increase by 2 units
d. decrease by 2 units

31. The correct relationship between SST, SSR, and SSE is given by
a. SSR = SST + SSE
b. SSR = SST - SSE
c. SSE = SSR - SST
d. None of these alternatives is correct.

32. In a multiple regression analysis SSR = 1,000 and SSE = 200. The F statistic for this model is
a. 5.0
b. 1,200
c. 800
d. Not enough information is provided to answer this question.

33. The ratio of MSE/MSR yields
a. SST
b. the F statistic
c. SSR
d. None of these alternatives is correct.

34. In a multiple regression analysis involving 12 independent variables and 166 observations, SSR = 878 and SSE = 122. The coefficient of determination is
    a. 0.1389
    b. 0.1220
    c. 0.878
    d. 0.7317

35. A regression analysis involved 8 independent variables and 99 observations. The critical value of t for testing the significance of each of the independent variable's coefficients will have
    a. 98 degrees of freedom
    b. 97 degrees of freedom
    c. 90 degrees of freedom
    d. 7 degrees of freedom

36. In order to test for the significance of a regression model involving 14 independent variables and 255 observations, the numerator and denominator degrees of freedom (respectively) for the critical value of F are
    a. 14 and 255
    b. 255 and 14
    c. 13 and 240
    d. 14 and 240

37. A multiple regression model has the form

    $$\hat{Y}=5+6X+7W$$

    As X increases by 1 unit (holding W constant), Y is expected to
    a. increase by 11 units
    b. decrease by 11 units
    c. increase by 6 units
    d. decrease by 6 units

38. In a multiple regression analysis involving 10 independent variables and 81 observations, SST = 120 and SSE = 42. The coefficient of determination is
    a. 0.81
    b. 0.11
    c. 0.35
    d. 0.65

39. For a multiple regression model, SST = 200 and SSE = 50. The multiple coefficient of determination is
    a. 0.25
    b. 4.00
    c. 250
    d. 0.75

40. A regression model involved 18 independent variables and 200 observations. The critical value of t for testing the significance of each of the independent variable's coefficients will have
    a. 18 degrees of freedom
    b. 200 degrees of freedom
    c. 199 degrees of freedom
    d. 181 degrees of freedom

41. In order to test for the significance of a regression model involving 8 independent variables and 121 observations, the numerator and denominator degrees of freedom (respectively) for the critical value of F are
    a. 8 and 121
    b. 7 and 120
    c. 8 and 112
    d. 7 and 112

42. In a multiple regression analysis involving 5 independent variables and 30 observations, SSR = 360 and SSE = 40. The coefficient of determination is
    a. 0.80
    b. 0.90
    c. 0.25
    d. 0.15

43. A regression analysis involved 6 independent variables and 27 observations. The critical value of t for testing the significance of each of the independent variable's coefficients will have
    a. 27 degrees of freedom
    b. 26 degrees of freedom
    c. 21 degrees of freedom
    d. 20 degrees of freedom

44. In order to test for the significance of a regression model involving 4 independent variables and 36 observations, the numerator and denominator degrees of freedom (respectively) for the critical value of F are
    a. 4 and 36
    b. 3 and 35
    c. 4 and 31
    d. 4 and 32

NARRBEGIN: Exhibit 15-1
Exhibit 15-1
In a regression model involving 44 observations, the following estimated regression equation was obtained.
$\hat{Y} = 29 + 18X_1 + 43X_2 + 87X_3$
For this model SSR = 600 and SSE = 400.
NARREND

45. Refer to Exhibit 15-1. The coefficient of determination for the above model is
    a. 0.667
    b. 0.600
    c. 0.336
    d. o.400

46. Refer to Exhibit 15-1. MSR for this model is
    a. 200
    b. 10
    c. 1,000
    d. 43

47. Refer to Exhibit 15-1. The computed F statistics for testing the significance of the above model is
    a. 1.500
    b. 20.00

c. 0.600
d. 0.6667

NARRBEGIN: Exhibit 15-2
Exhibit 15-2
A regression model between sales (Y in \$1,000), unit price (X_1 in dollars) and television advertisement (X_2 in dollars) resulted in the following function:

$\hat{Y} = 7 - 3X_1 + 5X_2$

For this model SSR = 3500, SSE = 1500, and the sample size is 18.
NARREND

48. Refer to Exhibit 15-2. The coefficient of the unit price indicates that if the unit price is
    a. increased by \$1 (holding advertising constant), sales are expected to increase by \$3
    b. decreased by \$1 (holding advertising constant), sales are expected to decrease by \$3
    c. increased by \$1 (holding advertising constant), sales are expected to increase by \$4,000
    d. increased by \$1 (holding advertising constant), sales are expected to decrease by \$3,000

49. Refer to Exhibit 15-2. The coefficient of X_2 indicates that if television advertising is increased by \$1 (holding the unit price constant), sales are expected to
    a. increase by \$5
    b. increase by \$12,000
    c. increase by \$5,000
    d. decrease by \$2,000

50. Refer to Exhibit 15-2. To test for the significance of the model, the test statistic F is
    a. 2.33
    b. 0.70
    c. 17.5
    d. 1.75

51. Refer to Exhibit 15-2. To test for the significance of the model, the *p*-value is
    a. less than 0.01
    b. between 0.01 and 0.025
    c. between 0.025 and 0.05
    d. between 0.05 and 0.10

52. Refer to Exhibit 15-2. The multiple coefficient of correlation for this problem is
    a. 0.70
    b. 0.8367
    c. 0.49
    d. 0.2289

NARRBEGIN: Exhibit 15-3
Exhibit 15-3
In a regression model involving 30 observations, the following estimated regression equation was obtained:
$\hat{Y} = 17 + 4X_1 - 3X_2 + 8X_3 + 8X_4$
For this model SSR = 700 and SSE = 100.
NARREND

53. Refer to Exhibit 15-3. The coefficient of determination for the above model is approximately
 a. -0.875
 b. 0.875
 c. 0.125
 d. 0.144

54. Refer to Exhibit 15-3. The computed F statistic for testing the significance of the above model is
 a. 43.75
 b. 0.875
 c. 50.19
 d. 7.00

55. Refer to Exhibit 15-3. The critical F value at 95% confidence is
 a. 2.53
 b. 2.69
 c. 2.76
 d. 2.99

56. Refer to Exhibit 15-3. The conclusion is that the
 a. model is not significant
 b. model is significant
 c. slope of X_1 is significant
 d. slope of X_2 is significant

NARRBEGIN: Exhibit 15-4
Exhibit 15-4

a. $Y=\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$
b. $E(Y)=\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$
c. $\hat{Y}=b_0+b_1X_1+b_2X_2$
d. $E(Y)=\beta_0 + \beta_1 X_1 + \beta_2 X_2$
NARREND

57. Which equation describes the multiple regression model?
 a. Equation A
 b. Equation B
 c. Equation C
 d. Equation D

58. Which equation gives the estimated regression line?
 a. Equation A
 b. Equation B
 c. Equation C
 d. Equation D

59. Which equation describes the multiple regression equation?
 a. Equation A
 b. Equation B
 c. Equation C
 d. Equation D

NARRBEGIN: Exhibit 15-5
Exhibit 15-5
Below you are given a partial Minitab output based on a sample of 25 observations.

	Coefficient	Standard Error
Constant	145.321	48.682
X_1	25.625	9.150
X_2	-5.720	3.575
X_3	0.823	0.183

NARREND

60. Refer to Exhibit 15-5. The estimated regression equation is
   a. $Y=\beta_0+\beta_1 X_1+\beta_2 X_2+\beta_3 X_3+\varepsilon$
   b. $E(Y)=\beta_0+\beta_1 X_1+\beta_2 X_2+\beta_3 X_3$
   c. $\hat{Y}=145.321+25.625X_1-5.720X_2+0.823X_3$
   d. $\hat{Y}=48.682+9.15X_1+3.575X_2+1.183X_3$

61. Refer to Exhibit 15-5. The interpretation of the coefficient on X_1 is that
   a. a one unit change in X_1 will lead to a 25.625 unit change in Y
   b. a one unit change in X_1 will lead to a 25.625 unit increase in Y when all other variables are held constant
   c. a one unit change in X_1 will lead to a 25.625 unit increase in X_2 when all other variables are held constant
   d. It is impossible to interpret the coefficient.

62. Refer to Exhibit 15-5. We want to test whether the parameter β_1 is significant. The test statistic equals
   a. 0.357
   b. 2.8
   c. 14
   d. 1.96

63. Refer to Exhibit 15-5. The t value obtained from the table to test an individual parameter at the 5% level is
   a. 2.06
   b. 2.069
   c. 2.074
   d. 2.080

64. Refer to Exhibit 15-5. Carry out the test of significance for the parameter β_1 at the 5% level. The null hypothesis should be
   a. rejected
   b. not rejected
   c. revised
   d. None of these alternatives is correct.

NARRBEGIN: Exhibit 15-6
Exhibit 15-6
Below you are given a partial computer output based on a sample of 16 observations.

	Coefficient	Standard Error
Constant	12.924	4.425
X_1	-3.682	2.630
X_2	45.216	12.560

Analysis of Variance

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression		4,853	2,426.5	
Error			485.3	

NARREND

65. Refer to Exhibit 15-6. The estimated regression equation is
   a. $Y=\beta_0+\beta_1X_1+\beta_2X_2+\varepsilon$
   b. $E(Y)=\beta_0+\beta_1X_1+\beta_2X_2$
   c. $\hat{Y}=12.924-3.682X_1+45.216X_2$
   d. $\hat{Y}=4.425+2.63X_1+12.56X_2$

66. Refer to Exhibit 15-6. The interpretation of the coefficient of X_1 is that
   a. a one unit change in X_1 will lead to a 3.682 unit decrease in Y
   b. a one unit increase in X_1 will lead to a 3.682 unit decrease in Y when all other variables are held constant
   c. a one unit increase in X_1 will lead to a 3.682 unit decrease in X_2 when all other variables are held constant
   d. It is impossible to interpret the coefficient.

67. Refer to Exhibit 15-6. We want to test whether the parameter β_1 is significant. The test statistic equals
   a. -1.4
   b. 1.4
   c. 3.6
   d. 5

68. Refer to Exhibit 15-6. The t value obtained from the table which is used to test an individual parameter at the 1% level is
   a. 2.65
   b. 2.921
   c. 2.977
   d. 3.012

69. Refer to Exhibit 15-6. Carry out the test of significance for the parameter β_1 at the 1% level. The null hypothesis should be
   a. rejected
   b. not rejected
   c. revised
   d. None of these alternatives is correct.

70. Refer to Exhibit 15-6. The degrees of freedom for the sum of squares explained by the regression (SSR) are
   a. 2
   b. 3
   c. 13
   d. 15

71. Refer to Exhibit 15-6. The sum of squares due to error (SSE) equals
   a. 37.33
   b. 485.3
   c. 4,853
   d. 6,308.9

72. Refer to Exhibit 15-6. The test statistic used to determine if there is a relationship among the variables equals
    a. -1.4
    b. 0.2
    c. 0.77
    d. 5

73. Refer to Exhibit 15-6. The F value obtained from the table used to test if there is a relationship among the variables at the 5% level equals
    a. 3.41
    b. 3.63
    c. 3.81
    d. 19.41

74. Refer to Exhibit 15-6. Carry out the test to determine if there is a relationship among the variables at the 5% level. The null hypothesis should
    a. be rejected
    b. not be rejected
    c. revised
    d. None of these alternatives is correct.

NARRBEGIN: Exhibit 15-7
Exhibit 15-7
A regression model involving 4 independent variables and a sample of 15 periods resulted in the following sum of squares.
SSR = 165
SSE = 60

NARREND

75. Refer to Exhibit 15-7. The coefficient of determination is
    a. 0.3636
    b. 0.7333
    c. 0.275
    d. 0.5

76. Refer to Exhibit 15-7. If we want to test for the significance of the model at 95% confidence, the critical F value (from the table) is
    a. 3.06
    b. 3.48
    c. 3.34
    d. 3.11

77. Refer to Exhibit 15-7. The test statistic from the information provided is
    a. 2.110
    b. 3.480
    c. 4.710
    d. 6.875

NARRBEGIN: Exhibit 15-8
Exhibit 15-8

The following estimated regression model was developed relating yearly income (Y in $1,000s) of 30 individuals with their age (X_1) and their gender (X_2) (0 if male and 1 if female).

$$\hat{Y}=30+0.7X_1+3X_2$$

Also provided are SST = 1,200 and SSE = 384.
NARREND

78. Refer to Exhibit 15-8. From the above function, it can be said that the expected yearly income of
    a. males is $3 more than females
    b. females is $3 more than males
    c. males is $3,000 more than females
    d. females is $3,000 more than males

79. Refer to Exhibit 15-8. The yearly income of a 24-year-old female individual is
    a. $19.80
    b. $19,800
    c. $49.80
    d. $49,800

80. Refer to Exhibit 15-8. The yearly income of a 24-year-old male individual is
    a. $13.80
    b. $13,800
    c. $46,800
    d. $49,800

81. Refer to Exhibit 15-8. The multiple coefficient of determination is
    a. 0.32
    b. 0.42
    c. 0.68
    d. 0.50

82. Refer to Exhibit 15-8. If we want to test for the significance of the model, the critical value of F at 95% confidence is
    a. 3.33
    b. 3.35
    c. 3.34
    d. 2.96

83. Refer to Exhibit 15-8. The test statistic for testing the significance of the model is
    a. 0.73
    b. 1.47
    c. 28.69
    d. 5.22

84. Refer to Exhibit 15-8. The model
    a. is significant
    b. is not significant
    c. would be significant is the sample size was larger than 30
    d. None of these alternatives is correct.

85. Refer to Exhibit 15-8. The estimated income of a 30-year-old male is
    a. $51,000
    b. $5,100
    c. $510
    d. $51

PROBLEM

1. The prices of Rawlston, Inc. stock (y) over a period of 12 days, the number of shares (in 100s) of company's stocks sold (x_1), and the volume of exchange (in millions) on the New York Stock Exchange (x_2) are shown below.

Day	(y)	(x1)	(x2)
1	87.50	950	11.00
2	86.00	945	11.25
3	84.00	940	11.75
4	83.00	930	11.75
5	84.50	935	12.00
6	84.00	935	13.00
7	82.00	932	13.25
8	80.00	938	14.50
9	78.50	925	15.00
10	79.00	900	16.50
11	77.00	875	17.00
12	77.50	870	17.50

Excel was used to determine the least-squares regression equation. Part of the computer output is shown below.

ANOVA

	df	SS	MS	F	Significance F
Regression	2	118.8474	59.4237	40.9216	0.0000
Residual	9	13.0692	1.4521		
Total	11	131.9167			

	Coefficients	Standard Error	t Stat	P-value
Intercept	118.5059	33.5753	3.5296	0.0064
(x1)	-0.0163	0.0315	-0.5171	0.6176
(x2)	-1.5726	0.3590	-4.3807	0.0018

a. Use the output shown above and write an equation that can be used to predict the price of the stock.
b. Interpret the coefficients of the estimated regression equation that you found in Part a.
c. At 95% confidence, determine which variables are significant and which are not.
d. If in a given day, the number of shares of the company that were sold was 94,500 and the volume of exchange on the New York Stock Exchange was 16 million, what would you expect the price of the stock to be?

2. Multiple regression analysis was used to study how an individual's income (Y in thousands of dollars) is influenced by age (X_1 in years), level of education (X_2 ranging from 1 to 5), and the person's gender (X_3 where 0 =female and 1=male). The following is a partial result of a computer program that was used on a sample of 20 individuals.

	Coefficient	Standard Error
X_1	0.6251	0.094
X_2	0.9210	0.190
X_3	-0.510	0.920

Analysis of Variance

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression		84		
Error		112		

a. Compute the coefficient of determination.
b. Perform a t test and determine whether or not the coefficient of the variable "level of education" (i.e., X_2) is significantly different from zero. Let $\alpha = 0.05$.
c. At $\alpha = 0.05$, perform an F test and determine whether or not the regression model is significant.
d. As you note the coefficient of X_3 is -0.510. Fully interpret the meaning of this coefficient.

3. A multiple regression analysis between yearly income (Y in $1,000s), college grade point average (X_1), age of the individuals (X_2), and the gender of the individual (X_3; zero representing female and one representing male) was performed on a sample of 10 people, and the following results were obtained.

	Coefficients	*Standard Error*
Intercept	4.0928	1.4400
X_1	10.0230	1.6512
X_2	0.1020	0.1225
X_3	-4.4811	1.4400

ANOVA

	DF	*SS*	*MS*	*F*
Regression		360.59		
Residual (Error)		23.91		

a. Write the regression equation for the above.
b. Interpret the meaning of the coefficient of X_3.
c. Compute the coefficient of determination.
d. Is the coefficient of X_1 significant? Use $\alpha = 0.05$.
e. Is the coefficient of X_2 significant? Use $\alpha = 0.05$.
f. Is the coefficient of X_3 significant? Use $\alpha = 0.05$.
g. Perform an F test and determine whether or not the model is significant.

4. The following results were obtained from a multiple regression analysis.

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression		384	48	
Error			20	
Total		704		

a. How many independent variables were involved in this model?
b. How many observations were involved?
c. Determine the F statistic.

5. Shown below is a partial computer output from a regression analysis.

	Coefficient	Standard Error
Constant	10.00	2.00
X_1	-2.00	1.50
X_2	6.00	2.00
X_3	-4.00	1.00

Analysis of Variance

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression		60		
Error				
Total	19	140		

a. Use the above results and write the regression equation.
b. Compute the coefficient of determination and fully interpret its meaning.
c. At $\alpha = 0.05$, test to see if there is a relation between X_1 and Y.
d. At $\alpha = 0.05$, test to see if there is a relation between X_3 and Y.
e. Is the regression model significant? Perform an F test and let $\alpha = 0.05$.

6. In order to determine whether or not the sales volume of a company (Y in millions of dollars) is related to advertising expenditures (X_1 in millions of dollars) and the number of salespeople (X_2), data were gathered for 10 years. Part of the regression results is shown below.

	Coefficients	*Standard Error*
Intercept	7.0174	1.8972
X_1	8.6233	2.3968
X_2	0.0858	0.1845

ANOVA

	DF	*SS*	*MS*	*F*
Regression		321.11		
Residual (Error)		63.39		

a. Use the above results and write the regression equation that can be used to predict sales.
b. Estimate the sales volume for an advertising expenditure of 3.5 million dollars and 45 salespeople. Give your answer in dollars.
c. At $\alpha = 0.05$, test to determine if the fitted equation developed in Part a represents a significant relationship between the independent variables and the dependent variable.
d. At $\alpha = 0.05$, test to see if β_1 is significantly different from zero.
e. Determine the multiple coefficient of determination.
f. Compute the adjusted coefficient of determination.

7. In order to determine whether or not the number of automobiles sold per day (Y) is related to price (X_1 in \$1,000), and the number of advertising spots (X_2), data were gathered for 7 days. Part of the regression results is shown below.

	Coefficient	Standard Error
Constant	0.8051	
X_1	0.4977	0.4617
X_2	0.4733	0.0387

Analysis of Variance

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression		40.700		
Error		1.016		

a. Determine the least squares regression function relating Y to X_1 and X_2.
b. If the company charges \$20,000 for each car and uses 10 advertising spots, how many cars would you expect them to sell in a day?
c. At $\alpha = 0.05$, test to determine if the fitted equation developed in Part a represents a significant relationship between the independent variables and the dependent variable.
d. At 95% confidence, test to see if price is a significant variable.
e. At 95% confidence, test to see if the number of advertising spots is a significant variable.
f. Determine the multiple coefficient of determination.

8. The following is part of the results of a regression analysis involving sales (Y in millions of dollars), advertising expenditures (X_1 in thousands of dollars), and number of salespeople (X_2) for a corporation. The regression was performed on a sample of 10 observations.

	Coefficient	Standard Error
Constant	-11.340	20.412
X_1	0.798	0.332
X_2	0.141	0.278

a. Write the regression equation.
b. Interpret the coefficients of the estimated regression equation found in Part (a).
c. At α =0.05, test for the significance of the coefficient of advertising.
d. At α =0.05, test for the significance of the coefficient of number of salespeople.
e. If the company uses $50,000 in advertisement and has 800 salespersons, what are the expected sales? Give your answer in dollars.

9. The following is part of the results of a regression analysis involving sales (Y in millions of dollars), advertising expenditures (X_1 in thousands of dollars), and number of sales people (X_2) for a corporation:

ANALYSIS OF VARIANCE

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Regression	2	822.088		
Error	7	736.012		

a. At α = 0.05 level of significance, test to determine if the model is significant. That is, determine if there exists a significant relationship between the independent variables and the dependent variable.
b. Determine the multiple coefficient of determination.
c. Determine the adjusted multiple coefficient of determination.
d. What has been the sample size for this regression analysis?

10. Below you are given a partial computer output based on a sample of 12 observations relating the number of personal computers sold by a computer shop per month (Y), unit price (X_1 in $1,000) and the number of advertising spots (X_2) used on a local television station.

	Coefficient	Standard Error
Constant	17.145	7.865
X_1	-0.104	3.282
X_2	1.376	0.250

a. Use the output shown above and write an equation that can be used to predict the monthly sales of computers.
b. Interpret the coefficients of the estimated regression equation found in Part a.
c. If the company charges $2,000 for each computer and uses 10 advertising spots, how many computers would you expect them to sell?
d. At α = 0.05, test to determine if the price is a significant variable.
e. At α = 0.05, test to determine if the number of advertising spots is a significant variable.

11. Below you are given a partial computer output based on a sample of 12 observations relating the number of personal computers sold by a computer shop per month (Y), unit price (X_1 in $1,000) and the number of advertising spots (X_2) they used on a local television station.

ANOVA

	DF	*SS*	*MS*	*F*	*Significance F*
Regression	2	655.955			
Residual	9				
Total		838.917			

a. At $\alpha = 0.05$ level of significance, test to determine if the model is significant. That is, determine if there exists a significant relationship between the independent variables and the dependent variable.
b. Determine the multiple coefficient of determination.
c. Determine the adjusted multiple coefficient of determination.

12. Below you are given a computer output based on a sample of 30 days of the price of a company's stock (Y in dollars), the Dow Jones industrial average (X_1), and the stock price of the company's major competitor (X_2 in dollars).

	Coefficient	Standard Error
Constant	20.000	5.455
X_1	0.006	0.002
X_2	-0.70	0.200

a. Use the output shown above and write an equation that can be used to predict the price of the stock.
b. If the Dow Jones Industrial Average is 10,000 and the price of the competitor is $50, what would you expect the price of the stock to be?
c. At $\alpha = 0.05$, test to determine if the Dow Jones average is a significant variable.
d. At $\alpha = 0.05$, test to determine if the stock price of the major competitor is a significant variable.

13. Below you are given a partial computer output relating the price of a company's stock (Y in dollars), the Dow Jones industrial average (X_1), and the stock price of the company's major competitor (X_2 in dollars).

ANOVA

	DF	*SS*	*MS*	*F*
Regression				
Residual	20	40		
Total		800		

a. What has been the sample size for this regression analysis?
b. At $\alpha = 0.05$ level of significance, test to determine if the model is significant. That is, determine if there exists a significant relationship between the independent variables and the dependent variable.
c. Determine the multiple coefficient of determination.

14. A regression was performed on a sample of 16 observations. The estimated equation is $\hat{Y} = 23.5 - 14.28X_1 + 6.72X_2 + 15.68X_3$. The standard errors for the coefficients are $S_{b1} = 4.2$, $S_{b2} = 5.6$, and $S_{b3} = 2.8$. For this model, SST = 3809.6 and SSR = 3285.4.
    a. Compute the appropriate t ratios.
    b. Test for the significance of β_1, β_2 and $\beta_3$at the 5% level of significance.
    c. Do you think that any of the variables should be dropped from the model? Explain.
    d. Compute R^2 and R_a^2. Interpret R^2.
    e. Test the significance of the relationship among the variables at the 5% level of significance.

15. The following results were obtained from a multiple regression analysis of supermarket profitability. The dependent variable, Y, is the profit (in thousands of dollars) and the independent variables, X_1 and X_2, are the food sales and nonfood sales (also in thousands of dollars).

    ANOVA

	DF	*SS*
Regression	2	562.363
Residual	9	225.326
Total		

	Coefficients	*Standard Error*
Intercept	-15.0621	
X_1	0.0972	0.054
X_2	0.2484	0.092

    a. Write the estimated regression equation for the relationship between the variables.
    b. Compute the coefficient of determination and fully interpret its meaning.
    c. Carry out a test of whether Y is significantly related to the independent variables. Use a 0.05 level of significance.
    d. Carry out a test to determine if there is a significant relationship between X_1 and Y. Use a .05 level of significance.
    e. How many supermarkets were in the sample used here?

16. A regression was performed on a sample of 20 observations. Two independent variables were included in the analysis, X and Z. The relationship between X and Z is $Z = X^2$. The following estimated equation was obtained.

    $\hat{Y} = 23.72 + 12.61X + 0.798Z$

    The standard errors for the coefficients are $S_{b1} = 4.85$ and $S_{b2} = 0.21$

    For this model, SSR = 520.2 and SSE = 340.6
    a. Estimate the value of Y when X = 5.
    b. Compute the appropriate t ratios.
    c. Test for the significance of the coefficients at the 5% level. Which variable(s) is (are) significant?
    d. Compute the coefficient of determination and the adjusted coefficient of determination. Interpret the meaning of the coefficient of determination.
    e. Test the significance of the relationship among the variables at the 5% level of significance.

17. A student used multiple regression analysis to study how family spending (Y) is influenced by income (X_1), family size (X_2), and additions to savings (X_3). The variables Y, X_1, and X_3 are measured in thousands of dollars. The following results were obtained.

ANOVA

	DF	*SS*
Regression	3	45.9634
Residual	11	2.6218
Total		

	Coefficients	*Standard Error*
Intercept	0.0136	
X_1	0.7992	0.074
X_2	0.2280	0.190
X_3	-0.5796	0.920

a. Write out the estimated regression equation for the relationship between the variables.
b. Compute R^2. What can you say about the strength of this relationship?
c. Carry out a test of whether Y is significantly related to the independent variables. Use a .05 level of significance.
d. Carry out a test to see if X_3 and Y are significantly related. Use a .05 level of significance.

18. A regression model involving 3 independent variables for a sample of 20 periods resulted in the following sum of squares.

	Sum of Squares
Regression	90
Residual (Error)	100

a. Compute the coefficient of determination and fully explain its meaning.
b. At $\alpha = 0.05$ level of significance, test to determine whether or not there is a significant relationship between the independent variables and the dependent variable.

19. A regression model involving 8 independent variables for a sample of 69 periods resulted in the following sum of squares.
SSE = 306
SST = 1800
a. Compute the coefficient of determination.
b. At $\alpha = 0.05$, test to determine whether or not the model is significant.

20. In a regression model involving 46 observations, the following estimated regression equation was obtained.
$\hat{Y} = 17 + 4X_1 - 3X_2 + 8X_3 + 5X_4 + 8X_5$
For this model, SST = 3410 and SSE = 510.
a. Compute the coefficient of determination.
b. Perform an F test and determine whether or not the regression model is significant.

21. A microcomputer manufacturer has developed a regression model relating his sales (Y in \$10,000s) with three independent variables. The three independent variables are price per unit (Price in \$100s), advertising (ADV in \$1,000s) and the number of product lines (Lines). Part of the regression results is shown below.

	Coefficient	Standard Error
Intercept	1.0211	22.8752
Price	-0.1524	0.1411
ADV	0.8849	0.2886
Lines	-0.1463	1.5340

Analysis of Variance

Source of Variation	Degrees of Freedom	Sum of Squares
Regression		2708.61
Error (Residuals)	14	2840.51

a. Use the above results and write the regression equation that can be used to predict sales.
b. If the manufacturer has 10 product lines, advertising of $40,000, and the price per unit is $3,000, what is your estimate of their sales? Give your answer in dollars.
c. Compute the coefficient of determination and fully interpret its meaning.
d. At $\alpha = 0.05$, test to see if there is a significant relationship between sales and unit price.
e. At $\alpha = 0.05$, test to see if there is a significant relationship between sales and the number of product lines.
f. Is the regression model significant? (Perform an F test.)
g. Fully interpret the meaning of the regression (coefficient of price) per unit that is, the slope for the price per unit.
h. What has been the sample size for this analysis?

22. The following is part of the results of a regression analysis involving sales (Y in millions of dollars), advertising expenditures (X_1 in thousands of dollars), and number of salespeople (X_2) for a corporation. The regression was performed on a sample of 10 observations.

	Coefficient	Standard Error
Intercept	40.00	7.00
X_1	8.00	2.50
X_2	6.00	3.00

a. If the company uses $40,000 in advertisement and has 30 salespersons, what are the expected sales? Give your answer in dollars.
b. At $\alpha = 0.05$, test for the significance of the coefficient of advertising.
c. At $\alpha = 0.05$, test for the significance of the coefficient of the number of salespeople.

23. Sherri Cola Company has developed a regression model relating its sales (Y in $10,000s) with four independent variables. The four independent variables are price per unit (PRICE, in dollars), competitor's price (COMPRICE, in dollars), advertising (ADV, in $1,000s) and type of container used (CONTAIN; 1 = Cans and 0 = Bottles). Part of the regression results is shown below. (Assume n = 25)

	Coefficient	Standard Error
Intercept	443.143	
PRICE	-57.170	20.426
COMPRICE	27.681	19.991
ADV	0.025	0.023
CONTAIN	-95.353	91.027

a. If the manufacturer uses can containers, his price is $1.25, advertising $200,000, and his competitor's price is $1.50, what is your estimate of his sales? Give your answer in dollars.
b. Test to see if there is a significant relationship between sales and unit price. Let $\alpha = 0.05$.
c. Test to see if there is a significant relationship between sales and advertising. Let $\alpha = 0.05$.
d. Is the type of container a significant variable? Let $\alpha = 0.05$.
e. Test to see if there is a significant relationship between sales and competitor's price. Let $\alpha = 0.05$.

24. The Brock Juice Company has developed a regression model relating sales (Y in \$10,000s) with four independent variables. The four independent variables are price per unit (X_1, in dollars), competitor's price (X_2, in dollars), advertising (X_3, in \$1,000s) and type of container used (X_4) (1 = Cans and 0 = Bottles). Part of the regression results are shown below:

Analysis of Variance

Source of Variation	Degrees of Freedom	Sum of Squares
Regression	4	283,940.60
Error (Residuals)	18	621,735.14

a. Compute the coefficient of determination and fully interpret its meaning.
b. Is the regression model significant? Explain what your answer implies. Let $\alpha = 0.05$.
c. What has been the sample size for this analysis?

25. The following regression model has been proposed to predict sales at a furniture store.

$$\hat{Y} = 10 - 4X_1 + 7X_2 + 18X_3$$

where

X_1=competitor's previous day's sales (in\$1,000s)

X_2=population within 1 mile (in 1000s)

X_3=1 if any form of advertising was used, 0 if otherwise

$\hat{Y}$=sales (in \$1,000s)

a. Fully interpret the meaning of the coefficient of X_3.
b. Predict sales (in dollars) for a store with competitor's previous day's sale of \$3,000, a population of 10,000 within 1 mile, and six radio advertisements.

26. A sample of 30 houses that were sold in the last year was taken. The value of the house (Y) was estimated. The independent variables included in the analysis were the number of rooms (X_1), the size of the lot (X_2), the number of bathrooms (X_3), and a dummy variable (X_4), which equals 0 if the house does not have a garage and equals 1 otherwise. The following results were obtained:

	Coefficients	*Standard Error*
Intercept	15,232.5	8,462.5
X_1	2,178.4	778.0
X_2	7.8	2.2
X_3	2,675.2	2,229.3
X_4	1,157.8	463.1

Analysis of Variance

	DF	*SS*	*MS*
Regression		204,242.88	51,060.72
Residual (Error)		205,890.00	8,235.60

a. Write out the estimated equation.
b. Interpret the coefficient on the number of rooms (X_1).
c. Interpret the coefficient on the dummy variable (X_4).
d. What are the degrees of freedom for the sum of squares explained by the regression (SSR) and the sum of squares due to error (SSE)?
e. Test whether or not there is a significant relationship between the value of a house and the independent variables. Use a .05 level of significance. Be sure to state the null and alternative hypotheses.
f. Test the significance of β_1 at the 5% level. Be sure to state the null and alternative hypotheses.

g. Compute the coefficient of determination and interpret its meaning.
h. Estimate the value of a house that has 9 rooms, a lot with an area of 7,500, 2 bathrooms, and 2 garages.

27. A sample of 25 families was taken. The objective of the study was to estimate the factors that determine the monthly expenditure on food for families. The independent variables included in the analysis were the number of members in the family (X_1), the number of meals eaten outside the home (X_2), and a dummy variable (X_3) that equals 1 if a family member is on a diet and equals 0 if there is no family member on a diet. The following results were obtained.

	Coefficients	*Standard Error*
Intercept	450.08	53.6
X_1	49.92	9.6
X_2	10.12	2.2
X_3	-.60	12.0

Analysis of Variance

	DF	*SS*	*MS*
Regression		3,078.39	1,026.13
Residual (Error)		2,013.90	95.90

a. Write out the estimated regression equation.
b. Interpret all coefficients.
c. Test for the significance of β_1, β_2, and β_3 at the 1% level of significance.
d. What are the degrees of freedom for the sum of squares explained by the regression (SSR) and the sum of squares due to error (SSE)?
e. Test whether of not there is a significant relationship between the monthly expenditure on food and the independent variables. Use a .01 level of significance. Be sure to state the null and alternative hypotheses.
f. Compute the coefficient of determination and explain its meaning.
g. Estimate the monthly expenditure on food for a family that has 4 members, eats out 3 times, and does not have any member of the family on a diet.

28. The following regression model has been proposed to predict sales at a fast food outlet.

$\hat{Y}=18-2X_1+7X_2+15X_3$

where

X_1=the number of competitors within 1 mile

X_2=the population within 1 mile (in 1000s)

X_3=1 if drive-up windows are present, 0 otherwise

$\hat{Y}$=sales (in$1,000s)

a. What is the interpretation of 15 (the coefficient of X_3) in the regression equation?
b. Predict sales for a store with 2 competitors, a population of 10,000 within one mile, and one drive-up window (give the answer in dollars).
c. Predict sales for the store with 2 competitors, a population of 10,000 within one mile, and no drive-up window (give the answer in dollars).

29. The following regression model has been proposed to predict sales at a computer store.

$$\hat{Y} = 50 - 3X_1 + 20X_2 + 10X_3$$

where

X_1 = competitor's previous day's sales (in $1,000s)

X_2 = population within 1 mile (in 1,000s)

$$X_3 = \begin{cases} 1 & \text{if radio advertising was used} \\ 0 & \text{if otherwise} \end{cases}$$

$\hat{Y}$ = sales (in $1,000s)

Predict sales (in dollars) for a store with the competitor's previous day's sale of $5,000, a population of 20,000 within 1 mile, and nine radio advertisements.

30. The following regression model has been proposed to predict monthly sales at a shoe store.

$$\hat{Y} = 40 - 3X_1 + 12X_2 + 10X_3$$

where

X_1 = competitor's previous month's sales (in $1,000s)

X_2 = store's previous month's sales (in $1,000s)

$$X_3 = \begin{cases} 1 & \text{if radio advertising was used} \\ 0 & \text{if otherwise} \end{cases}$$

$\hat{Y}$ = sales (in $1,000s)

a. Predict sales (in dollars) for the shoe store if the competitor's previous month's sales were $9,000, the store's previous month's sales were $30,000, and no radio advertisements were run.
b. Predict sales (in dollars) for the shoe store if the competitor's previous month's sales were $9,000, the store's previous month's sales were $30,000, and 10 radio advertisements were run.